

AYMANE AIT DADS

Data Science & AI Intern | Machine Learning · Data Analysis · Python

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PROFESSIONAL SUMMARY

Data Science engineering student (EURECOM) who automated network quality classification across 100K+ fiber-optic records at Orange Maroc, cutting manual analysis time by 60% and directly informing commercial service-tier decisions for the network optimization team. Ranked top 10% on a \$106K+ Kaggle competition leaderboard (0.80/1.0) by fine-tuning a 30B Mamba-Transformer MoE model with LoRA on chain-of-thought reasoning tasks, demonstrating end-to-end AI model development under production GPU constraints. Core competencies span Python, SQL, machine learning, deep learning, NLP, computer vision, and data storytelling through Power BI dashboards and executive-facing presentations. Available for internship in Europe from Summer 2025; holds valid residence in France with no sponsorship requirement.

CORE COMPETENCIES

Machine Learning | Data Analysis | Artificial Intelligence | Python Programming | Natural Language Processing | Computer Vision | Data Storytelling | Model Evaluation

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and delivered **Power BI dashboards and stakeholder presentations** translating Random Forest and K-Means model outputs into actionable network maintenance recommendations for the optimization team.
- Built an automated clustering pipeline that mapped customer performance profiles across **100K+ fiber-optic network records** (upload speed, latency, jitter) to 5 commercial service tiers used directly in production.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and end-to-end pipeline development in Python and Scikit-learn.
- Classified network quality across thousands of nodes using ensemble machine learning methods, producing structured outputs ingested by the **network optimization team** for infrastructure planning.

PROJECTS

Aerial Cactus Detection - Endangered Species Classification

EURECOM · 2025 · 99.8% Accuracy | F1 = 0.999 | ROC AUC = 1.0

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images**, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0 for endangered species classification.
- Conducted systematic model evaluation and ablation studies comparing architecture families, producing clear performance reports mapping model outputs to a real-world conservation data pipeline.

PyTorch · DenseNet121 · CNN · Scikit-learn · Kaggle GPU

Twitter Sentiment Analysis - End-to-End NLP Pipeline

EURECOM · 2025 · Production NLP Pipeline

- Constructed a full NLP pipeline progressing from tokenization and TF-IDF through word2vec embeddings to **transformer fine-tuning**, optimizing each stage for sentiment classification accuracy.
- Applied systematic hyperparameter optimization across model variants, generating data-driven insights and evaluation reports that translate model performance into actionable classification decisions.

Python · Hugging Face Transformers · Scikit-learn · Pandas · NumPy

NVIDIA Nemotron Reasoning Challenge - Kaggle Competition

Kaggle · In Progress (June 2026) · Top 10% | \$106K+ Prize Pool

- Achieved **top 10% public leaderboard (0.80/1.0)** on a \$106K+ prize-pool competition by fine-tuning a 30B Mamba-Transformer MoE model using LoRA (r=32, ~888M trainable parameters) with SFTTrainer on 7,828 chain-of-thought reasoning examples.
- Resolved 5 Blackwell GPU (sm_120) incompatibilities in strict dependency order and executed 20+ single-variable ablations - confirming completion-only loss, packing strategy, and adapter scale as key performance drivers.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Data Science & Analytics: Python | SQL | Pandas | NumPy | Power BI | Scikit-learn | Feature Engineering

Machine Learning & AI: PyTorch | Hugging Face Transformers | NLP | Computer Vision | Ensemble Methods | LoRA | Clustering

MLOps & Infrastructure: Git | Linux | Kaggle GPU | FP16 Mixed Precision | Docker | REST APIs | Ablation Studies